Student Performance Predictor & Management System

📌 Overview of the Project

The **Student Performance Predictor & Management System** is a software solution designed to analyze student academic data and predict performance outcomes using data-driven models. It allows educators, administrators, and institutions to track academic progress, identify weak-performing students early, and make informed decisions to improve overall learning outcomes.

The system also functions as a management tool for storing academic records, generating analysis reports, visualizations, and performance insights.

⭐ Features

- Predicts students' academic performance using machine learning models
- Stores student profiles and academic records
- User-friendly dashboard for visualization and reports
- Automated alerts for academically at-risk students
- Supports data uploads in spreadsheet format
- Interactive performance charts and analytics
- Role-based access (Admin, Faculty)
- Light-weight UI for easy navigation

🛠 Technologies / Tools Used

- **Programming Language:** Python
- **Frameworks & Libraries:**
  - Flask / Django (backend)
  - Pandas, NumPy
  - Scikit-Learn (ML algorithms)
  - Matplotlib / Seaborn / Plotly (visualizations)
- **Frontend:** HTML, CSS, JavaScript
- **Database:** SQLite / MySQL
- **Environment:** Jupyter Notebook / VS Code

⚙ Steps to Install & Run the Project

1️⃣ Clone the Repository

```
git clone https://github.com/your-repository-url.git

cd student-performance-system
```

2️⃣ Create a Virtual Environment (optional but recommended)

```
python -m venv venv

source venv/bin/activate   # Linux/Mac

venv\Scripts\activate     # Windows
```

3️⃣ Install Dependencies

```
pip install -r requirements.txt
```

4️⃣ Run Database Migrations (if any)

```
python manage.py migrate
```

5️⃣ Start the Server

```
python manage.py runserver
```

Once the server is running, open the browser and visit:

http://localhost:8000

🧪 Instructions for Testing

- Login using test credentials (if provided in seed data)
- Upload a dataset containing student records
- Run predictive analysis and observe model output
- Check system dashboards for:
  - Predictions
  - Data visualizations
  - Student risk analysis
- Validate accuracy through real academic results if available

```
====== STUDENT MANAGEMENT SYSTEM ======
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 3
Enter Student ID to search: s001

Student Found:
ID: s001
Name: Pratim
Age: 19
Course: BTech

====== STUDENT MANAGEMENT SYSTEM ======
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 2

--- Student Records ---
ID: s001
Name: Pratim
Age: 19
Course: BTech
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====== STUDENT MANAGEMENT SYSTEM ======
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1
Enter Student ID: s001
Enter Student Name: Pratim
Enter Age: 19
Enter Course: BTech
Student added successfully!
```

```
====== STUDENT MANAGEMENT SYSTEM ======
1. Add Student
2. View All Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 6
Exiting program. Goodbye!
```